T14: 3D Affordance Prediction

Team 7 | 8-minute talk script + 2-minute Q&A guide

How to use this

The main talk covers slides 1-12. Slides 13-14 are backup slides for questions. This script is written for two speakers; swap the speaker labels if needed. Speak naturally and use the timestamps as checkpoints. You do not need to read every slide word-for-word.

Timing checkpoints

Slides 1-5: Speaker 1, about 2:55. Slides 6-12: Speaker 2, about 5:05. Total: 8:00, followed by the professor's 2-minute Q&A.

Spoken script

SLIDE 1 | 0:00-0:30 | SPEAKER 1

Hello, we are Team 7. Our topic is 3D affordance prediction: finding which parts of an object support an action. We will explain the task, what we learned from our experiments, and the next experiment we propose. Our focus is how the wording of an action query affects the predicted region.

SLIDE 2 | 0:30-1:00 | SPEAKER 1

The input is a 3D point cloud and an action query, such as "grasp" or "press." The model assigns a score or label to every point, so different queries can highlight different parts of the same object. This is difficult when the object is only partly visible, because the point cloud may not include the part that matters.

SLIDE 3 | 1:00-1:45 | SPEAKER 1

Our baseline is the released OpenAD PointNet++ checkpoint. We kept the weights fixed and did not retrain the model. In our adapted E01 evaluation, partial-view mean IoU was 12.50 percent, and the full-shape result was 14.41 percent. E05 later evaluated all 9,140 partial views and got 12.54 percent, but it used a different batch and point-sampling setup, so we report it separately. We also profiled one 2,048-point pass on our M3 Pro using PyTorch MPS. That timing and memory reading describe a separate profile, not the E01 benchmark run.

Spoken script | continued

SLIDE 4 | 1:45-2:25 | SPEAKER 1

We compared query names while keeping the checkpoint and point clouds the same. With training-name queries, partial-view mean IoU across 18 actions was 40.39 percent. With held-out query names, it was 10.77 percent. The scores also dropped for full shapes. This shows that predictions are sensitive to query wording. It does not prove generalization to entirely new action concepts.

SLIDE 5 | 2:25-2:55 | SPEAKER 1

We also looked at a failure on one partial vase. The target labels marked 563 points as supporting "pour," but the model predicted none of those points correctly. On those target points, its average score for "none" was much higher than its score for "pour." This is one example, not a dataset-wide result. A partial view may hide useful cues, but we have not tested that as the cause.

SLIDE 6 | 2:55-3:40 | SPEAKER 2

Next, we tested whether a different phrase could help. In the OpenAD prompt-transfer pilot, one preselected paraphrase improved "unlock" IoU from 3.69 percent with the canonical word to 12.36 percent. But the negative-object any-alarm rate also rose from 0.34 to 1.61 percent, and the other four tested prompts did not show a similar gain. These were exploratory validation results, not an official test-set result. So one phrase can help, but it may also create more false alarms.

SLIDE 7 | 3:40-4:25 | SPEAKER 2

Our earlier prompt experiments used an unofficial Espresso-3D reimplementation with Affogato-pretrained weights. They were small exploratory studies, not official Affogato benchmark results. The effects varied by action, and averaging prompts often lowered scores. One grab result improved, but threshold experiments also showed that reducing false alarms could leave us with no positive detections. We have not run Affostruction inference. Together, these experiments suggest that prompt choice needs a controlled evaluation.

Spoken script | research plan

SLIDE 8 | 4:25-5:10 | SPEAKER 2

Our research question is: across 18 actions, how do one paraphrase and an average of three paraphrases change localization and false alarms compared with canonical action words? Our hypothesis is that effects will vary by action and that averaging three prompts will not give a reliable overall IoU gain. Any gains may come with more false alarms. We will compare the three-prompt average with the canonical words as our primary test. If the paired 95 percent confidence interval is entirely above zero, that would go against our prediction.

SLIDE 9 | 5:10-5:45 | SPEAKER 2

To make the comparison fair, we will keep the model and point samples fixed. We will compare three conditions: the canonical word, one preregistered paraphrase, and the average of three fixed paraphrase score maps. Partial views are our primary setting, with full shapes as a control. We will report mean IoU, per-action results, AUC, and false alarms on negative objects, with paired confidence intervals resampled by object.

SLIDE 10 | 5:45-6:25 | SPEAKER 2

First, we will write the paraphrases and freeze them before evaluation. Then we will run all three conditions on the same inputs. We will measure the primary IoU comparison and track false alarms and action-level changes. Finally, we will check the full-shape control and inspect failure cases. A fresh object-disjoint test set would be best. If we have to reuse validation data, we will describe the result as exploratory.

SLIDE 11 | 6:25-7:35 | SPEAKER 2

Our schedule runs from Week 9 to Week 15. In Week 9, we will finalize the question, primary comparison, and data split. In Week 10, we will write and freeze the prompts. In Week 11, we will check the scoring and averaging script on a small sample. In Week 12, we will run the partial-view comparison. In Week 13, we will run the full-shape control and calculate per-action results and false alarms. In Week 14, we will calculate paired confidence intervals and inspect failures. In Week 15, we will summarize the findings and limitations. Throughout, we will keep the model, objects, and point samples consistent.

SLIDE 12 | 7:35-8:00 | SPEAKER 2

To conclude, our experiments show that query wording can change OpenAD predictions, but improvements are inconsistent and can increase false alarms. Our next experiment will test prompt strategies under controlled conditions. Thank you. We are happy to take your questions.

Two-minute Q&A: answer guide

Keep answers short: answer directly, give one piece of evidence, then state the limitation. These are the questions most likely to come up.

What is your baseline?

We use the released OpenAD PointNet++ checkpoint with its weights frozen. Our E01 results come from an adapted evaluation, so we do not claim it is an untouched reproduction of the official benchmark.

Why do E01 and E05 have slightly different scores?

They used different batch and point-sampling conditions. E01 partial mIoU was 12.50 percent; E05 all-view, point-weighted mIoU was 12.54 percent. We report them separately.

What do you mean by held-out query names?

We changed the query vocabulary while keeping the model and point clouds the same. The result shows sensitivity to wording, not proof of generalization to a completely new action concept.

Why test paraphrases if earlier results were mixed?

Because the results were mixed, we do not assume paraphrases help. We will control the comparison and measure both IoU and false alarms.

How will you judge the hypothesis?

Our primary comparison is the three-prompt average versus canonical words. A paired confidence interval fully above zero would contradict our prediction. If it crosses zero, the result is inconclusive.

Did you run Affogato and Affostruction?

We ran small exploratory tests with an unofficial Espresso-3D reimplementation using Affogato-pretrained weights. We did not run an official Affogato benchmark reproduction, and we have not run Affostruction inference.

Does an affordance score mean a robot will succeed?

No. It is a model score for how well a point matches an affordance query. It does not directly measure whether a real robot can complete the action.

What is your main limitation?

Some prompt studies are exploratory and use validation data or small samples. A fresh object-disjoint test set and fixed prompts would make the next study stronger.

If you do not know an answer

Say: "We have not tested that directly yet. In our proposed experiment, we will measure it by..." Do not guess, and do not present exploratory results as official benchmark results.